

UNIVERSITY OF MALAYSIA TERENGGANU

FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

CSF3143 BASICS OF SOFTWARE ENGINEERING

GROUP: 1

TITLE:

SMART RESIDENTIAL COLLEGE LAUNDRY MANAGEMENT SYSTEM

LECTURER:

TS. MOHAMMAD AIZAT BIN BASIR

STUDENTS' NAME AND MATRICS NUMBER:

HAFFNIS BINTI MAHADI (S87507)

TAN SHEN YI (S87271)

HOCK CHENG TAO (S87348)

NADJATUN NAJIHAH BINTI MANSOR (S88178)

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1. Project Background

1.1 Problem Statement and Justification

The current laundry facilities at Universiti Malaysia Terengganu (UMT) residential colleges face several problems that affect students' laundry experience. Students often have to visit the laundry room just to check whether washing machines and dryers are available. During busy periods, finding an available machine can be difficult and may result in unnecessary waiting time. Furthermore, some students leave their clothes inside machines even after their washing cycles have been completed, causing inconvenience to other users.

Based on the questionnaire conducted among 23 students, 78.3% of respondents usually check machine availability by visiting the laundry room directly, while 65.2% have experienced situations where previous users left their clothes inside washing machines. Interviews with UMT Residential College Student Representative Council (JPKK UMT) and laundry supervisors also highlighted problems involving faulty machines, delayed maintenance reports and the lack of a proper system to monitor machine usage.

Therefore, the Smart Residential College Laundry Management System is proposed to improve the existing laundry management process. The system aims to provide students with easier access to machine availability, online reservations, queue management and washing completion notifications. It will also help laundry supervisors monitor machine usage, manage faulty machine reports and plan maintenance more effectively. Through this system, laundry activities can become more organised, convenient and efficient for students and laundry management.

1.2 Proposed System Overview

1.2.1 System Introduction

The proposed system is called the Smart Residential College Laundry Management System. It is designed to improve the laundry management process at Universiti Malaysia Terengganu (UMT) residential colleges by making laundry activities more convenient, organised, and efficient.

Through this system, students can check the availability of washing machines and dryers, reserve available washing machines based on their preferred time slots, and receive notifications when their washing cycles are completed. This helps students save time and manage their laundry schedules more effectively. Moreover, laundry supervisors can monitor machine usage records and review faulty machine reports to plan maintenance and servicing accordingly.

1.2.2 System Objectives

The objectives of the proposed system are:

1. To allow students to check the availability and status of washing machines and dryers.
2. To provide a reservation function that allows students to book available washing machines and dryers based on their preferred time slots.
3. To provide an online queue and notify students when their washing and drying cycles are completed.
4. To allow students to identify machines with completed laundry cycles but uncollected clothes.
5. To allow students to report damaged or malfunctioning washing machines and dryers.
6. To help laundry supervisors monitor machine usage records and plan maintenance schedules accordingly.

1.2.3 Main System Modules

The proposed system consists of six main modules:

1. Machine Availability Checking – Allows students to check whether washing machines and dryers are available, occupied, or under maintenance.

2. Laundry Reservation – Allows students to reserve available washing machines and dryers according to their preferred time slots.

3. Laundry Completion Notification and Queue Management – Allows students to join an online queue and receive notifications when their washing and drying cycles are completed.

4. Completed Laundry Status – Allows students to identify machines that have completed their washing and drying cycles but still contain uncollected clothes.

5. Faulty Machine Reporting – Allows students to report damaged or malfunctioning washing machines and dryers through the system.

6. Machine Usage Monitoring and Maintenance – Allows laundry supervisors to monitor the usage frequency of washing machines and dryers and use the records to plan maintenance and servicing schedules.

1.2.4 System Users

The proposed system involves two main types of users:

1. Student
Students can check machine availability, make laundry reservations, receive washing completion notifications, check completed laundry status, and report faulty machines.

2. Laundry Supervisors
The laundry supervisors are responsible for monitoring machine conditions, reviewing faulty machine reports, and updating machine statuses to ensure the information in the system remains accurate.

1.2.5 System Scope and Limitations

The proposed system focuses on managing laundry facilities in UMT residential colleges through a mobile application and/or website. It covers machine availability checking, laundry reservations, washing and drying completion notifications, completed laundry status, faulty machine reporting, and machine usage monitoring.

However, the existing token-based payment mechanism will remain unchanged. The system will not include online payment or token purchasing functions. In addition, the accuracy of machine availability, washing progress and completion status depends on timely updates by students or laundry supervisors. The system does not automatically detect machine conditions or

communicate directly with washing machines and dryers. The system is also limited to laundry management and will not cover other residential college facilities.

1.3 Software Development Process

The Agile Software Development Model, specifically the **Scrum Framework**, is used for the development of the system. Its iterative development process, flexibility and emphasis on collaboration among team members make it suitable for the project's scope, team size and timeline. Scrum also allows the development team to continuously improve the system based on feedback and changing requirements.

1.3.1 Justification of Selection of Scrum Framework

1. Incremental System Development

   The Smart Residential College Laundry Management System consists of several modules, such as user authentication, machine availability checking, laundry reservations and digital queue management. Scrum allows these modules to be developed and tested incrementally through different Sprints, making the development process more manageable.

2. Continuous Testing and Feedback

   Our team plans to involve students staying in the residential colleges in the testing and evaluation process throughout each Sprint. This provides opportunities to obtain feedback on the developed functionalities and identify potential errors. Furthermore, feedback from students and laundry management can also be used to improve the usability and effectiveness of the system.

3. Improved Team Collaboration

   Scrum encourages regular communication and coordination among team members. Through Sprint Planning, progress discussions and Sprint Reviews, each member can understand their responsibilities and contribute towards achieving the project objectives.

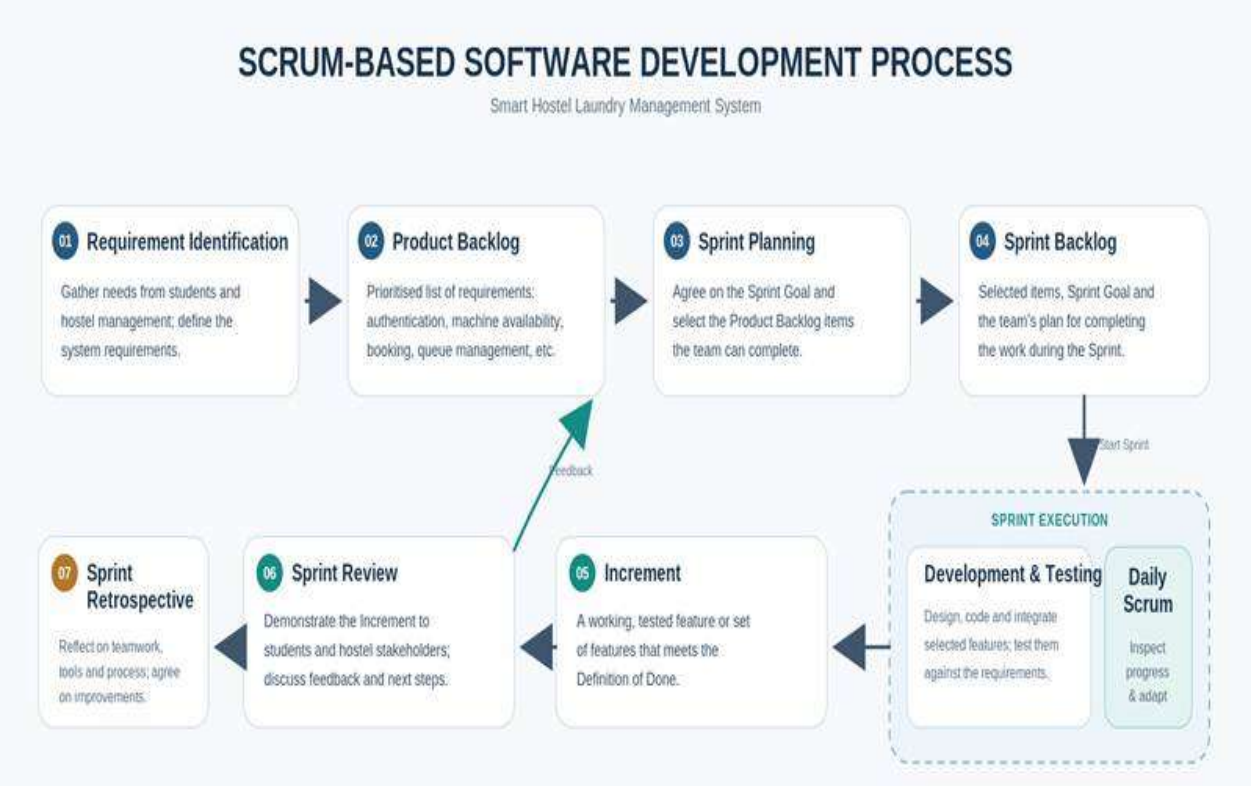


Figure 1.3 Scrum-Based Software Development Process

https://scrumguides.org/scrum-guide.html#purpose-of-the-scrum-guide

1.3.2 Description of Scrum-Based Software Development Process

The development process begins with requirement identification, where the team gathers and documents the necessary system requirements. These requirements are organized into a Product Backlog, which contains all the functionalities and improvements required for the system.

Next, Sprint Planning is conducted to determine which functionalities will be developed during the upcoming Sprint. The selected tasks are organized into a Sprint Backlog and assigned to the respective team members.

During Sprint Execution, team members develop, integrate and test the selected functionalities. Regular communication is maintained to monitor progress and resolve development issues.

At the end of each Sprint, a Sprint Review is conducted to demonstrate the completed functionalities and obtain feedback. Subsequently, a Sprint Retrospective is performed to identify areas for improvement in the development process.

The Product Backlog is then updated based on the feedback and remaining requirements before the next Sprint begins. This iterative process continues until the essential functionalities of the Smart Residential College Laundry Management System are completed.

1.4 Project Planning (Roles, GitHub & Timeline)

Our Smart Residential College Laundry Management System project is carried out by a group of four members in which each member is assigned specific responsibilities to ensure that all project tasks are completed systematically.

1.4.1 Members' Roles

The group members and their respective roles are shown in the table below.

	Haffnis binti Mahadi (S87507) Scrum Master Project and stakeholder coordinator She is responsible for coordinating project activities, monitoring the project timeline, organising group discussion and managing communication with stakeholders.
	Hock Cheng Tao (S87348) Developer System design and technical coordinator He is responsible for coordinating technical planning, identifying non-functional requirements and contributes to the development of the system design, use case diagram and user interface.

	TAN SHEN YI (S87271) Developer System analyst and requirement coordinator He is responsible for analysing stakeholder needs, identifying system functions, preparing functional requirements and ensuring that the proposed system meets users' needs.
	Nadjatun Najihah binti Mansor (S88178) Developer Research and quality assurance coordinator She is responsible for assisting in identifying project problems, reviewing the accuracy and completeness of requirements and checking the quality of project documentation.

All members are also responsible for participating in project discussions, reviewing project progress and contributing to completion of the final project.

1.4.2 Collaboration Platform

GitHub is used as the main platform to document projects and collaboration. A repository was created to organise and store project materials. All four group members were added as collaborators to enable shared access and participate in project activities.

The GitHub repository serves as the main platform for storing project documents, sharing updates and managing different versions of the project. Group discussions are also conducted through WhatsApp to ease the communication and decision-making among members. Important discussions or announcements were updated for future reference. The group plans to use GitHub to manage project tasks, record progress, store project documents and maintain updated versions of the project throughout the development process.

1.4.3 Project Timeline

The project timeline is prepared to ensure that all the project activities are organised and completed according to the given schedule. It also helps to monitor project progress and complete each milestone within the timeframe given.

Date	Week	Project Activities
22 October 2026	Week 3	Project formation, project title selection and GitHub repository creation
7 November 2026	Week 5	Milestone 1: Requirement and Use Case Model
19 November 2026	Week 7	Milestone 2: System Design
24 December 2026	Week 12	Final project submission
3-7 January 2027	Week 14	Project presentation

The group will follow the planned timeline to ensure that project tasks are completed systematically and all deliverables are submitted on time.

2. System Requirements

2.1 Stakeholder Identification and Requirements Gathering

2.1.1 Stakeholder Identification

The main stakeholders involved in the existing laundry operation and the proposed system are:

1. Students – The main users of the laundry facilities who require information about machine availability, queue status and machine conditions.
2. Laundry Supervisors – The personnel responsible for monitoring laundry operations, handling maintenance issues and managing machine usage.
3. UMT Residential College Student Representative Council (JPKK UMT) – The representative body of the residential colleges that acts as a communication channel between students and the relevant laundry operators regarding facility-related issues.

2.1.2 Requirement Elicitation Techniques

Several requirement elicitation techniques were used to gather information about the existing laundry operations and identify users' needs. These techniques include:

- Interviews: Interviews were conducted with JPKK UMT representatives and laundry supervisors to understand the existing management process, common problems and system requirements.
- Observation: Observations were carried out to understand the current laundry environment, machine usage and difficulties faced by students.
- Questionnaire: A Google Forms questionnaire was distributed to students to collect feedback about their laundry experiences, common problems and opinions on the proposed system.

2.1.3 Findings from Requirement Elicitation

1. Questionnaire Findings

Based on the questionnaire conducted among 23 respondents through Google Forms, Figures A.1 to A.3 show that 78.3% of students usually check machine availability by visiting the laundry room directly. Meanwhile, 65.2% have experienced situations where previous users left their clothes inside washing machines after the washing cycle was completed, while 47.8% find it difficult to locate an available washing machine. These findings highlight the need for a system that improves laundry facility management and provides better access to machine availability information.

The proposed system also includes a feature that allows students to report faulty washing machines. According to the questionnaire results, 78.3% of respondents agreed that this feature would be useful, while 17.4% were unsure but showed some interest in using it. Only 4.3% disagreed. These results indicate that most respondents support the implementation of a faulty washing machine reporting system.

2. Interview with JPKK UMT

The interview with JPKK UMT revealed that the self-service laundry facilities in residential colleges are managed by either the UMT Student Cooperative or external laundry operators. Therefore, the actual operation and maintenance of laundry machines are mainly handled by the respective operators rather than the college administration.

JPKK UMT acts as a communication channel between students and the relevant laundry operators when facility-related problems are reported. Students usually report these problems through the contact numbers provided at each laundry room, and JPKK UMT forwards the complaints to the respective operators when necessary.

The interview also identified several common problems, including malfunctioning washing machines, unpleasant smells from washing machines and dryers that leave clothes damp. Furthermore, JPKK UMT agreed that a digital system displaying machine availability and laundry status would be useful. It could help students check machine conditions and reduce unnecessary waiting times.

3. Interview with Laundry Supervisors

The interview with the laundry supervisor, Mr. Mazlen, highlighted several operational problems. One of the most common complaints concerns payment issues, such as insufficient tokens and unsuccessful payment verification due to Internet connection problems. Nonetheless, this issue is not considered in the project scope as it is related to payment platforms that required high professionalism.

Currently, students report faulty machines through WhatsApp messages, phone calls or direct messages. However, some reports may be overlooked, causing delays in handling maintenance issues. Furthermore, there is no dedicated system for informing students about machine repair progress or whether a faulty machine is ready for use again.

The laundry supervisor also suggested several improvements for the proposed system:

- Machine Usage Records: Record the frequency of washing machine and dryer usage to help maintenance personnel monitor machine usage and plan maintenance activities.
- Student Authentication: Implement a login system using student accounts or real names to ensure that only legitimate students can use the queueing system.
- Live Washing Progress: Display the current laundry status, such as waiting, in progress and completed, to help students monitor their laundry activities.

2.1.4 Requirement Analysis and Validation

Based on the questionnaire, interviews and observations, several important system requirements were identified.

For students, the system should provide machine availability checking, laundry reservations, online queue management and completion notifications. It should also allow students to identify machines containing uncollected laundry and report faulty machines.

For laundry supervisors, the system should provide machine usage records, allow them to update machine information and repair statuses, and help them monitor reported problems. Student authentication is also required to ensure that only registered students can access the system's laundry management functions.

The identified requirements were analysed based on the feedback collected from students, JPKK UMT and laundry supervisors to ensure that the proposed system addresses the actual problems faced by its stakeholders.

2.2 Functional Requirements

Functional requirements describe the main functions and services that the Smart Residential College Laundry Management System must provide to its users. Based on the questionnaire results collected from 23 respondents, external laundry supervisor and JPKK UMT, several important requirements were identified. These requirements explain how the system helps students manage their laundry activities more efficiently and allows the management to monitor and maintain the laundry facilities.

The functional requirements of the proposed system are as follows:

1. User Registration and Login

- The system shall allow students to register an account using their personal information.
- The system shall allow registered users to log in using their username and password.
- The system shall provide different access permissions for students and laundry supervisors.
- The system shall allow laundry supervisors to access administrative functions using authorised accounts
- The system shall allow users to log out from their accounts.
- The system shall require students to log in using their registered accounts before accessing the laundry management system.

2. Machine Availability Checking

- The system shall display the availability status of washing machines and dryers.
- The system shall show whether a machine is available, occupied, or under maintenance.

- The system shall allow authorised laundry supervisors to update machine statuses whenever changes occur.
- The system shall allow students to identify machines with completed laundry cycles but uncollected clothes.

3. Laundry Reservation and Queue Management

- The system shall allow students to select an available washing machine and dryer.
- The system shall allow students to choose their preferred laundry date and time slot.
- The system shall prevent students from booking machines that are already reserved.
- The system shall allow students to join the laundry queue online without having to wait physically in line.
- The system shall allow students to check their current queue status through the system.
- The system shall display the number of students still waiting in the queue.

4. Laundry Status Tracking and Completion Notification

- The system shall display the current laundry status, including waiting, in progress and completed.
- The system shall allow students to identify machines with completed laundry cycles but uncollected clothes.
- The system shall send notifications to students when their washing and drying cycle is completed.
- The system shall remind students to collect their clothes on time after the washing and dryer cycle is completed.

5. Faulty Machine Reporting and Maintenance Management

- The system shall allow students to report damaged or malfunctioning washing machines and dryers.
- The system shall allow students to provide details about the problems or damage to the machines.

- The system shall allow the laundry supervisors to view and manage reported machine problems.
- The system shall allow management to update washing machine and dryer repair statuses so students can know whether a machine is still under maintenance or has been repaired.

- The system shall allow management to monitor machine usage records to assist in scheduling maintenance.

In conclusion, these functional requirements are designed based on the problems and suggestions identified through the questionnaires. The proposed functions focus on machine availability, machines usage tracking, laundry reservation, queue management, washing progress notifications and faulty machine reporting. These functions aim to improve the efficiency of laundry management, reduce students' waiting time and provide a more convenient laundry experience for students at UMT residential colleges.

2.3 Non-functional Requirements

Non-functional requirements describe the quality, performance, and overall characteristics that the Smart Residential College Laundry Management System should provide.
These requirements explain how well the system should operate and what users can expect when using it. They are important to ensure that the system is user-friendly, reliable, secure, accessible and easy to maintain. The following are the non-functional requirements identified for this system.

No.	Non-functional Requirement	Description
1	Usability	The system should provide a simple and user-friendly interface that allows students to easily understand and use its features.

2	Reliability	The system should operate consistently and provide accurate, up-to-date information to students. It should minimise system errors, prevent incorrect queue records, and handle unexpected interruptions properly to ensure a smooth and dependable laundry management process.
3	Security	The system should protect students' personal information and prevent unauthorized access through secure authentication and appropriate access permissions.
4	Availability	The system should be accessible whenever students need to check machine availability and queue status. It should provide updated information on machine conditions, such as available, in use, or out of service, while minimising system downtime.
5	Maintainability	The system should allow authorised administrators to update machine information, manage faulty machine reports and maintain system records easily.

2.4 Quality and Completeness of Requirements

The requirements gathering process for the Smart Residential College Laundry Management System was carried out to ensure that the proposed system features are relevant, clear and complete. The requirements were based on feedback collected through a Google Forms questionnaire involving 23 UMT students, as well as interviews with JPKK UMT representatives and the laundry supervisor, Mr. Mazlen. The collected feedback and identified operational problems were translated into structured functional and non-functional requirements.

To ensure completeness, the identified requirements cover the main activities and responsibilities of the system users, as follows:

1. Student Module Capabilities

- Registering and logging into the system.
- Checking washing machine and dryer availability.
- Reserving available machines and joining the online queue.
- Checking queue status and the number of waiting students.
- Monitoring laundry status and receiving completion notifications.
- Identifying machines containing completed but uncollected laundry.
- Reporting faulty or damaged machines.

2. Laundry Supervisor Capabilities

- Accessing administrative functions through authorised accounts.
- Updating machine availability and conditions.
- Monitoring machine usage records to assist in maintenance scheduling.
- Viewing and managing faulty machine reports.
- Updating machine repair and maintenance statuses.

3. Quality and System Requirements (Non-functional Requirements)

- User-friendly interface for students and laundry supervisors.
- Reliable system operations and accurate records.

- Secure authentication and appropriate access permissions.
- System accessibility with minimal downtime.
- Easy system maintenance and management.

Overall, the identified requirements cover the main functions needed to improve laundry management at UMT residential colleges while maintaining the existing token-based payment mechanism. These requirements provide a clear foundation for the subsequent system design and development.

5. Appendices

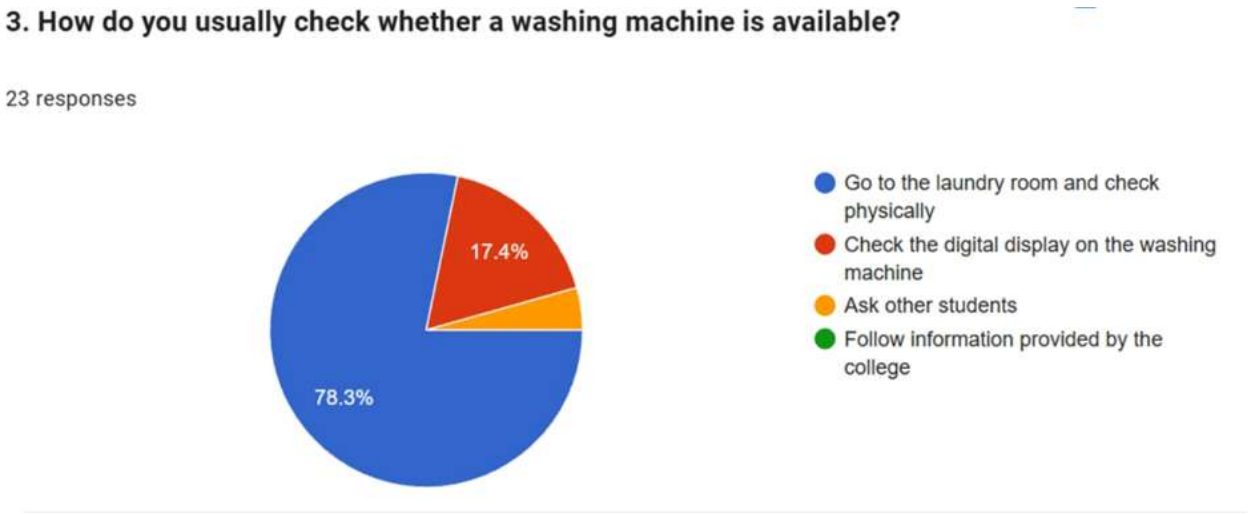


Figure A.1: *Survey Results on Preferred Platform for the Laundry Management System*

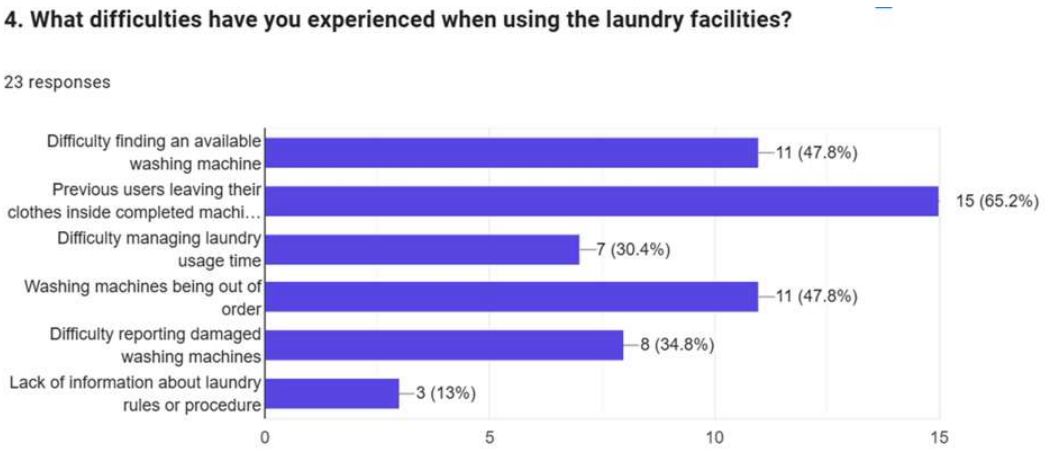


Figure A.2: *Survey Results on Difficulties Experienced When Using Laundry Facilities*

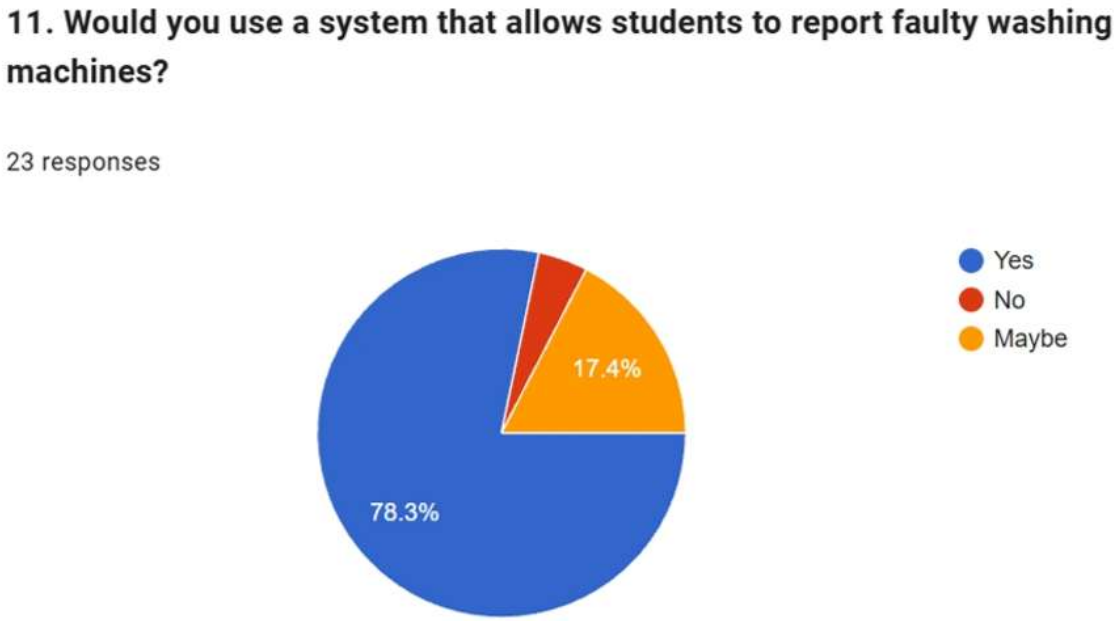


Figure A.3: *Students' Willingness to Use a System for Reporting Faulty Washing Machines*